Week 6 — Law of Large Numbers, Duality, Conic Programming

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Oct 3, 2026

1. Law of large numbers

$\tilde{\xi}_1, \tilde{\xi}_2, \tilde{\xi}_3, \ldots$: an infinite sequence of **independent and identically distributed (i.i.d.)** random variables, each with expected value μ

Law of large numbers (strong law, SLLN)

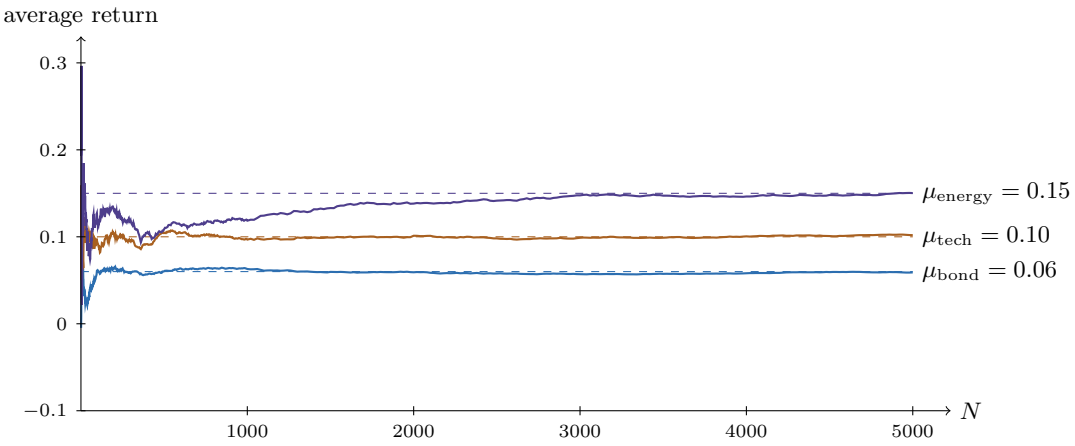
$$\mathbb{P}\Big(\lim_{N\to\infty} \frac{1}{N} \sum_{i\in[N]} \tilde{\xi}_i \;=\; \mu \Big) \;=\; 1$$

sample average $\longrightarrow$ expected value

Example portfolio: $\tilde{\xi}_i$ = return of an asset on day i, i.i.d. over days

asset	expected return μ	σ
bond	0.06	0.10
tech	0.10	0.20
energy	0.15	0.30

$\frac{1}{N} \sum_{i\in[N]} \tilde{\xi}_i$ = average return over the first N days



running sample average of each asset — each one settles at its own μ

2. Duality theory

in this course, we focus on **convex** functions.

Primal Problem

$$\begin{array}{ll} \min & f_0(x) \\ \text{s.t.} & x \in \mathbb{R}^N \\ & f_i(x) \le 0 \qquad \forall i \in [I] \end{array} \tag{P}$$

Dual Problem

$$\begin{array}{lll} \max\limits_{\theta} \ \min\limits_{x \in \mathbb{R}^N} & f_0(x) + \sum\limits_{i \in [I]} \theta_i f_i(x) \\ \text{s.t.} & \theta \in \mathbb{R}^I \\ & \theta \ge 0 \end{array} \tag{D}$$

$$\begin{array}{|l|} \hline \\ \begin{array}{ll} \min & f_0(x) \\ \text{s.t.} & x \in \mathbb{R}^N \\ & f_i(x) \le 0 \qquad \forall i \in [I] \end{array} \\ \hline \\ = \min\limits_{x \in \mathbb{R}^N} \ f_0(x) \ + \underbrace{\max\limits_{\theta \in \mathbb{R}^I_+} \sum\limits_{i \in [I]} \theta_i f_i(x)}_{\le 0} \\ \\ \hline \end{array}$$

$$\begin{array}{lll} \text{if} & f_i(x) \le 0 \ \ \forall i, & \text{then} \quad = 0 \\ \text{if} & \exists\, i : f_i(x) > 0, & \text{then} \quad = +\infty \end{array}$$

$$\ge \max_{\theta \in \mathbb{R}^I_+} \ \min_{x \in \mathbb{R}^N} \ f_0(x) + \sum_{i \in [I]} \theta_i f_i(x)$$

Properties:

— **weak duality** $\min(P) \ \ge \ \max(D)$ always holds.

— **strong duality:** $\min(P) \ = \ \max(D)$ holds

when the problem (P) is convex and the Slater's condition holds

$$\exists\, x \in \mathbb{R}^N, \ \text{s.t.} \ \ f_i(x) < 0 \quad \forall i \in [I]$$

In this course, we assume that strong duality holds for convex problem!!

3. Example: linear program

Primal LP:

$$\begin{array}{ll} \min & c^\top x \\ \text{s.t.} & x \in \mathbb{R}^N \\ & Ax \le b \end{array} \qquad \begin{array}{ll} f_0(x) = c^\top x & \\ f_i(x) = a_i^\top x - b_i & \forall i \in [I] \end{array}$$

$$\begin{aligned} & f_i(x) \le 0 \\ \iff & a_i^\top x - b_i \le 0 \\ \iff & a_i^\top x \le b_i \end{aligned}$$

Dual LP:

$$\max_{\theta \in \mathbb{R}^I_+} \min_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i \big(a_i^\top x - b_i\big)$$

$$= \max_{\theta \in \mathbb{R}^I_+} \min_{x \in \mathbb{R}^N} c^\top x + \sum_{i \in [I]} \theta_i a_i^\top x - \sum_{i \in [I]} \theta_i b_i$$

$$= \max_{\theta \in \mathbb{R}^I_+} -\sum_{i \in [I]} \theta_i b_i + \min_{x \in \mathbb{R}^N} \underbrace{\Big(c + \sum_{i \in [I]} \theta_i a_i\Big)^\top x}_{= \begin{cases} 0 & \text{if } c + \sum_{i \in [I]} \theta_i a_i = 0 \\ -\infty & \text{if otherwise.} \end{cases}}$$

$$\begin{array}{ll} = \max & -b^\top \theta \\ \text{s.t.} & \theta \in \mathbb{R}^I_+ \\ & A^\top \theta = -c \end{array}$$

LP strong duality : if there exists $x \in \mathbb{R}^N : Ax \le b$

or $\theta \in \mathbb{R}^I : A^\top \theta = -c, \quad \theta \ge 0$

then strong duality holds

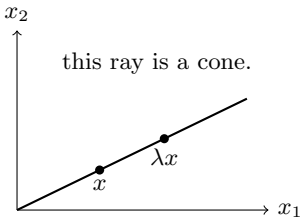
4. Conic programming

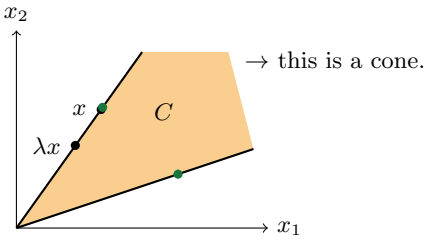
a conic program is a convex optimization problem.

— a **linear** objective function.

— a feasible set defines through the intersection of halfspaces $Ax \le b$ and a proper cone C.

Def (*cone*) a set C is a cone if $\forall x \in C$, we have $\lambda x \in C \quad \forall \lambda \ge 0$





Def (*proper cone*) $\mathbb{R}^N$ is a cone but not pointed.

C is a proper cone if it's **convex**, **closed**, **pointed**

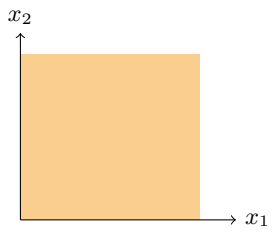
if $x, -x \in C$ then $x = 0$.

Generic conic program:

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \le b \\ & x \in C \end{aligned}$$

Example:

1) **Linear program:** $C = \mathbb{R}^N_+$

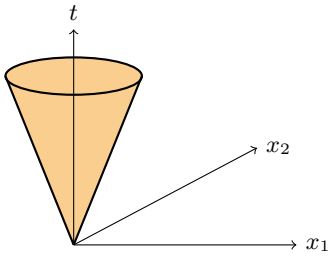


$$x \in \mathbb{R}^N_+ \iff x \ge 0$$

$$\min \; c^\top x \;\; \text{s.t.} \;\; Ax \le b, \;\; x \ge 0$$

2) **Second-order cone:** $C = \mathrm{SOC}^{N+1}_Q = \left\{(x,t) \in \mathbb{R}^N \times \mathbb{R} \;:\; \sqrt{x^\top Q\, x} \le t\right\}, \quad Q \succ 0$

$Q = I$: $\mathrm{SOC}^{N+1} = \left\{(x,t) \;:\; \|x\|_2 \le t\right\}$ "ice-cream" cone



3) **Semidefinite program:** $C = \mathbb{S}^K_+$

$\mathbb{S}^K$: set of symmetric matrices in $\mathbb{R}^{K\times K}$ $\quad (A^\top = A)$

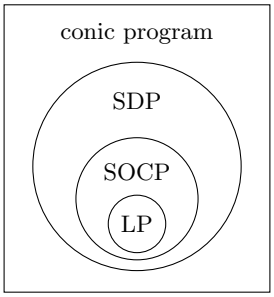
$\mathbb{S}^K_+$: set of positive semidefinite (symmetric) matrices in $\mathbb{R}^{K\times K}$

$A \in \mathbb{S}^K_+$ if $z^\top A z \geq 0 \qquad \forall z \in \mathbb{R}^K$

Generic semidefinite program: the variable is a matrix X

$$\begin{array}{ll} \min & \langle D, X\rangle \\ \text{s.t.} & X \in \mathbb{S}^K \\ & \langle A_i, X\rangle \leq b_i \qquad \forall i \in [I] \\ & X \in \mathbb{S}^K_+ \end{array} \qquad \rightarrow \text{usually ignored}$$

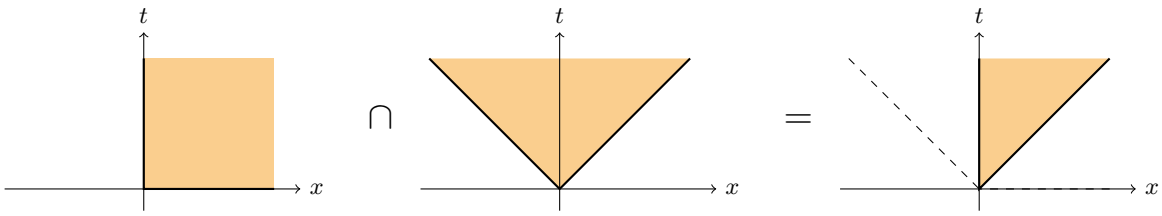
$\langle D, X\rangle = \sum_{j,k} D_{jk} X_{jk}$ — linear in X



4) **Several cones at once:**

$x \in C_1$ and $x \in C_2 \iff x \in C_1 \cap C_2$ — again a proper cone

$x = (x^1, x^2)$: $x^1 \in C_1$ and $x^2 \in C_2 \iff x \in C_1 \times C_2$ — again a proper cone



$C_1 = \mathbb{R}^2_+$ $\qquad$ $C_2 = \text{SOC}^2 :\ |x| \leq t$ $\qquad$ $C_1 \cap C_2 :\ 0 \leq x \leq t$

Example: portfolio, minimum variance

$$\begin{array}{lll} \min & x^\top \Sigma\, x & \\ \text{s.t.} & x \in \mathbb{R}^N & \\ & \mu^\top x \geq r & \iff \\ & e^\top x \leq 1 & \\ & x \geq 0 & \end{array} \qquad \begin{array}{lll} \min & t & \\ \text{s.t.} & (x,t) \in \mathbb{R}^N \times \mathbb{R} & \\ & \mu^\top x \geq r,\ \ e^\top x \leq 1 & \iff A\,(x,t) \leq b \qquad \text{halfspaces} \\ & \sqrt{x^\top \Sigma\, x} \leq t & \iff (x,\ t) \in \text{SOC}^{N+1}_\Sigma \qquad \text{cone} \\ & x \geq 0 & \iff (x,\ t) \in \mathbb{R}^{N+1}_+ \qquad \text{cone} \end{array}$$

$\Rightarrow\ C = \text{SOC}^{N+1}_\Sigma \cap \mathbb{R}^{N+1}_+$ — a SOCP in (x,t)

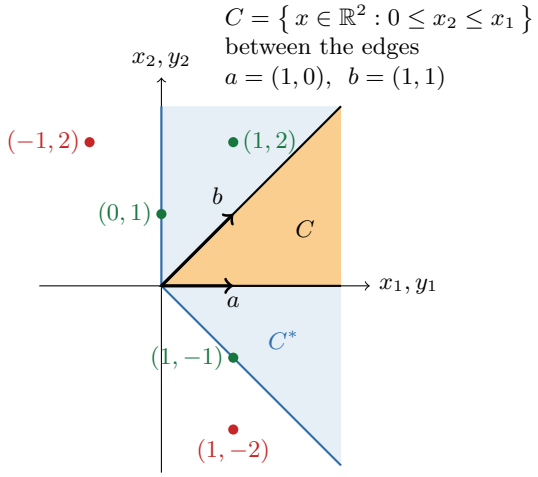
5. Conic programming duality

Def (*dual cone*)

$$C^* = \{\, y \in \mathbb{R}^n \;:\; y^\top x \geq 0 \quad \forall x \in C \,\}$$

$$C^* = \{\, y \in \mathbb{R}^2 : y_1 \geq 0, \;\; y_1 + y_2 \geq 0 \,\}$$

y	$y^\top a = y_1$	$y^\top b = y_1 + y_2$	$y \in C^*$?
$(1, 2)$	1	3	yes
$(0, 1)$	0	1	yes
$(1, -1)$	1	0	yes
$(-1, 2)$	-1	1	no
$(1, -2)$	1	-1	no



Example:

1) $C = \mathbb{R}^n_+ \;:\; x \geq 0 \quad \Longrightarrow \quad C^* = \mathbb{R}^n_+ = C$

   $n = 2$: edges $(1, 0)$, $(0, 1)$ $\Rightarrow$ $C^* = \{\, y : y_1 \geq 0, \; y_2 \geq 0 \,\} = C$

2) $C = \mathrm{SOC}^n \;:\; \|x\|_2 \leq t$ (second-order cone, $Q = I$) $\Longrightarrow \quad C^* = \mathrm{SOC}^n = C$

3) $C = \mathrm{SOC}^n_Q \;:\; \sqrt{x^\top Q\, x} \leq t, \quad Q \succ 0$

   $\Longrightarrow \quad C^* = \mathrm{SOC}^n_{Q^{-1}} \;:\; \sqrt{x^\top Q^{-1} x} \leq t$

$*$ **Note** $C = \mathbb{R}^n_+ = C^*$ and $C = \mathrm{SOC}^n = C^*$

Primal conic program:

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathbb{R}^N \\ & Ax \le b \\ & x \in C \end{aligned}$$

Dual conic program: dualize only $Ax \le b$; keep $x \in C$ in the inner min

$$\max_{\theta \in \mathbb{R}^I_+} \min_{x \in C} \; c^\top x + \sum_{i \in [I]} \theta_i \big(a_i^\top x - b_i\big) \qquad \sum_{i \in [I]} \theta_i a_i = A^\top \theta, \quad \sum_{i \in [I]} \theta_i b_i = b^\top \theta$$

$$= \max_{\theta \in \mathbb{R}^I_+} \; -b^\top \theta \; + \; \min_{x \in C} \underbrace{\big(c + A^\top \theta\big)^\top x}$$

$$= \begin{cases} 0 & \text{if } \; c + A^\top \theta \in C^* \\ -\infty & \text{otherwise} \end{cases}$$

$y = c + A^\top \theta$:

— $y \in C^*$: $\quad y^\top x \ge 0 \;\; \forall x \in C, \quad \text{and} \;\; x = 0 \in C \quad \Rightarrow \;\; \min = 0$

— $y \notin C^*$: $\quad \exists\, \bar{x} \in C : y^\top \bar{x} < 0, \quad \text{and} \;\; \lambda \bar{x} \in C \;\; \forall \lambda \ge 0 \;\; \text{(cone)}$

$$\Rightarrow \;\; y^\top (\lambda \bar{x}) = \lambda\, y^\top \bar{x} \to -\infty \;\; \text{as} \;\; \lambda \to \infty$$

$$\begin{aligned} = \; \max \quad & -b^\top \theta \\ \text{s.t.} \quad & \theta \in \mathbb{R}^I_+ \\ & c + A^\top \theta \in C^* \end{aligned}$$

Example 1: linear program, $C = \mathbb{R}^N_+ = C^*$

Primal		**Dual**	
min	$c^\top x$	max	$-b^\top \theta$
s.t.	$x \in \mathbb{R}^N$	s.t.	$\theta \in \mathbb{R}^I_+$
	$Ax \le b$		$c + A^\top \theta \in \mathbb{R}^N_+ = C^*$
	$x \in \mathbb{R}^N_+ = C$		

Example 2: quadratic program, $Q \succ 0$

$$\begin{array}{ll} \min & x^\top Q\, x \\ \text{s.t.} & x \in \mathbb{R}^N \\ & Ax \le b \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \min & t \\ \text{s.t.} & (x,t) \in \mathbb{R}^N \times \mathbb{R} \\ & Ax \le b \\ & \sqrt{x^\top Q\, x} \le t \end{array}$$

epigraph: same minimizer x^*, $\quad t^* = \sqrt{x^{*\top} Q\, x^*}$

decision $z = \begin{bmatrix} x \\ t \end{bmatrix} \in \mathbb{R}^{N+1}, \qquad c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad c^\top z = t$

$Ax \le b \iff \bar{A} z \le b, \quad \bar{A} = \begin{bmatrix} A & 0 \end{bmatrix} \qquad \sqrt{x^\top Q\, x} \le t \iff z \in \mathrm{SOC}_Q^{N+1}$

$$\begin{array}{ll} \min & c^\top z \\ \text{s.t.} & z \in \mathbb{R}^{N+1} \\ & \bar{A} z \le b \\ & z \in \mathrm{SOC}_Q^{N+1} = C \end{array} \qquad \text{— a SOCP}$$

Dual: dualize only $\bar{A} z \le b$; $\quad \theta \in \mathbb{R}^I_+$ — one entry per row of A

$$\max_{\theta \in \mathbb{R}^I_+} \min_{z \in C} c^\top z + \theta^\top \left(\bar{A} z - b \right)$$

$$= \max_{\theta \in \mathbb{R}^I_+} -b^\top \theta \; + \min_{z \in C} \left(c + \bar{A}^\top \theta \right)^\top z \qquad \Rightarrow \; c + \bar{A}^\top \theta \in C^* = \mathrm{SOC}_{Q^{-1}}^{N+1}$$

$$\bar{A}^\top \theta = \begin{bmatrix} A^\top \\ 0^\top \end{bmatrix} \theta = \begin{bmatrix} A^\top \theta \\ 0 \end{bmatrix}$$

$$c + \bar{A}^\top \theta = \begin{bmatrix} 0 \\ 1 \end{bmatrix} + \begin{bmatrix} A^\top \theta \\ 0 \end{bmatrix} = \begin{bmatrix} A^\top \theta \\ 1 \end{bmatrix} \in \mathrm{SOC}_{Q^{-1}}^{N+1} \iff \sqrt{(A^\top \theta)^\top\, Q^{-1}\, (A^\top \theta)} \le 1$$

it is just $\sqrt{y^\top Q^{-1} y} \le 1$ with $y = A^\top \theta$

Primal		**Dual**	
min	$c^\top z$	max	$-b^\top \theta$
s.t.	$z \in \mathbb{R}^{N+1}$	s.t.	$\theta \in \mathbb{R}^I_+$
	$\bar{A} z \le b$		$\theta^\top A\, Q^{-1} A^\top \theta \le 1$
	$z \in \mathrm{SOC}_Q^{N+1}$		

Remark $\quad Q = I$: $\quad \sqrt{x^\top x} = \|x\|_2 \le t \iff z \in \mathrm{SOC}^{N+1} = C = C^*, \qquad$ dual: $\quad \|A^\top \theta\|_2 \le 1$

Example 3: portfolio, minimum variance — Example 2 with $Q = \Sigma$

$$\begin{array}{ll} \min & x^\top \Sigma\, x \\ \text{s.t.} & x \in \mathbb{R}^N \\ & \mu^\top x \geq r \\ & e^\top x \leq 1 \\ & x \geq 0 \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \min & t \\ \text{s.t.} & (x,t) \in \mathbb{R}^N \times \mathbb{R} \\ & Ax \leq b \\ & (x,t) \in \mathrm{SOC}_\Sigma^{N+1} \end{array}$$

$\mu^\top x \geq r,\ \ e^\top x \leq 1,\ \ x \geq 0 \iff Ax \leq b$: $\quad A = \begin{bmatrix} -\mu^\top \\ e^\top \\ -I \end{bmatrix}, \quad b = \begin{bmatrix} -r \\ 1 \\ 0 \end{bmatrix}$

dual: $\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} \in \mathbb{R}_+^{N+2}, \quad \theta_1, \theta_2 \in \mathbb{R}_+, \ \ \theta_3 \in \mathbb{R}_+^N$ — one entry per row of A

$A^\top = \begin{bmatrix} -\mu & e & -I \end{bmatrix}, \qquad A^\top \theta = \theta_2 e - \theta_1 \mu - \theta_3, \qquad -b^\top \theta = r\,\theta_1 - \theta_2$

Primal

$$\begin{array}{ll} \min & t \\ \text{s.t.} & (x,t) \in \mathbb{R}^N \times \mathbb{R} \\ & Ax \leq b \\ & (x,t) \in \mathrm{SOC}_\Sigma^{N+1} \end{array}$$

Dual

$$\begin{array}{ll} \max & r\,\theta_1 - \theta_2 \\ \text{s.t.} & \theta_1, \theta_2 \geq 0, \ \ \theta_3 \in \mathbb{R}_+^N \\ & (\theta_2 e - \theta_1 \mu - \theta_3)^\top \Sigma^{-1} (\theta_2 e - \theta_1 \mu - \theta_3) \leq 1 \end{array}$$

$\Sigma = I$: $\quad \|\theta_2 e - \theta_1 \mu - \theta_3\|_2 \leq 1$